

GNN-BERT for Understanding Context from Music

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Abstract—This paper presents our implementation of a hybrid BERT and Graph Neural Network (GNN) system for understanding musical context, completed as a course project for CSE425 (Neural Networks). The project follows a four-task specification: a BERT-only text baseline, a GNN model over audio segment graphs, a GNN-BERT fusion model with a multi-task classification and emotion-regression objective, and a contrastive text-audio alignment task. We use the DEAM dataset as our primary source of audio and emotion labels, deriving four mood-quadrant tags from its continuous valence-arousal annotations, and combine MusicCaps and FMA-small for the contrastive alignment task. We report macro/micro F1 and area-under-precision-recall-curve for the classification tasks, and mean absolute error and R^2 for the emotion regression task. We also report and discuss a data-loading issue that prevented us from obtaining results for the contrastive alignment task at the time of writing.

Index Terms—graph neural networks, BERT, music information retrieval, multimodal fusion, music emotion recognition

I. INTRODUCTION

Understanding a piece of music computationally requires more than recognizing a single attribute such as genre. A listener’s impression of a track is shaped by several layers at once, including its harmonic and rhythmic structure, its emotional tone, and any text associated with it, such as tags or descriptions. This project, assigned as part of the CSE425 Neural Networks course, asked us to build a system that combines two complementary neural architectures to model this kind of context: a pretrained BERT [1] language model for text, and a Graph Neural Network (GNN) for the structural relationships between segments of an audio track.

The task specification, including the four sub-tasks described in this report, was provided by the course instructor as part of the assignment. Our team’s contributions were the choice of datasets used to implement the pipeline and the full implementation, training, and evaluation of all four tasks.

The remainder of this report is organized as follows. Section II describes the datasets we selected. Section III describes our preprocessing pipeline. Section IV describes the four tasks and their implementation. Section V reports our results. Section

VI discusses limitations, including an unresolved data-loading issue in Task 4. Section VII concludes the report.

II. DATASETS

We selected three datasets to satisfy the audio and text/tag requirements of the four tasks from the list of datasets that were specified. For the primary dataset we used DEAM. We also used FMA and MusicCaps datasets for this project.

A. DEAM (Primary Dataset)

The MediaEval Database for Emotional Analysis in Music (DEAM) [3] is our primary dataset, used for Tasks 1 through 3. It contains 1,802 song excerpts, each annotated with continuous valence and arousal scores on a 1–9 scale, representing how positive and how energetic a piece of music is perceived to be. Since DEAM does not provide discrete tags, we derived a four-class label for each track by dividing the valence-arousal plane into quadrants, following the same idea used by the related EmoMusic dataset: *happy/energetic*, *calm/content*, *sad/depressed*, and *angry/tense*. These quadrant labels serve two purposes in our pipeline: they are the classification target for Task 1 and Task 3, and the two words naming each quadrant (e.g. “happy energetic”) are used as the pseudo-caption text fed into BERT, since DEAM does not otherwise supply text descriptions. The original continuous valence and arousal values are retained as the regression target used in Task 3.

B. MusicCaps and FMA-small

For the contrastive alignment task, we combined two additional datasets. MusicCaps [4] provides natural-language captions for approximately 5,500 ten-second clips of audio sourced from YouTube; captions are the intended text side of the Task 4 contrastive pairs. FMA-small, drawn from the Free Music Archive dataset [5], provides audio with a single genre label per track, which we use as a weaker pseudo-caption to enlarge the pool of text-audio pairs available for contrastive training.

III. PREPROCESSING

Every audio clip is resampled to 22,050 Hz. We extract chroma features (12 bins) using `librosa`, then divide each clip into a fixed number of equal-length segments (eight in our implementation) and average the chroma vectors within each segment, giving one feature vector per segment. Each track is represented as a graph in which every node is a segment: edges connect temporally adjacent segments, and edges are also added between any two segments whose chroma vectors have a cosine similarity above a fixed threshold, so that segments which sound alike are linked even if they are far apart in time. This graph is the input to the GNN described in Section IV-B.

IV. METHODOLOGY

This section describes the four tasks, in the order specified by the assignment.

A. Task 1: BERT Baseline

Task 1 is a text-only multi-label classifier. The pseudocaption for each track (Section II-A) is passed through a pretrained `distilbert-base-uncased` model, and the embedding of the [CLS] token is passed through a single linear layer with a sigmoid activation to predict the four mood-quadrant tags. The model is trained with binary cross-entropy loss. This baseline has no access to audio at all, and serves as a reference point for how much the graph and fusion models improve on text alone.

B. Task 2: Graph Neural Network

Task 2 operates only on the segment graphs, with no text input. We implemented a simplified GraphSAGE-style [2] message-passing layer: for every node, we average the feature vectors of its neighbors (using the graph’s adjacency matrix), concatenate this neighbor average with the node’s own feature vector, and pass the result through a linear layer followed by a ReLU activation. Two such layers are stacked, and the final node representations are mean-pooled across all nodes in the graph to produce a single embedding per track, which is passed through a classifier for the same four-class tagging task as in Task 1. As required by the assignment, we compare this against a baseline that does not use the graph structure at all: a small two-layer convolutional network (CNN) applied directly to the segment-feature matrix, treated as a single-channel image.

C. Task 3: GNN-BERT Fusion

Task 3 combines the text embedding from Task 1’s BERT encoder with the graph embedding from Task 2’s GNN encoder. We implemented and compared two fusion strategies, in line with the ablation requested by the assignment:

- **Concatenation:** the two embeddings are concatenated and passed through a linear layer.
- **Attention:** the graph embedding is used to compute a scalar attention weight over the text embedding, which is then combined with the graph embedding.

In both variants, the fused representation feeds two output heads: a sigmoid classification head predicting the four mood-quadrant tags, and a second head predicting the continuous valence and arousal values. The model is trained with a combined loss,

$$\mathcal{L} = \mathcal{L}_{\text{tags}} + \alpha \mathcal{L}_{\text{emotion}}, \quad (1)$$

where $\mathcal{L}_{\text{tags}}$ is binary cross-entropy on the mood-quadrant tags, $\mathcal{L}_{\text{emotion}}$ is the mean squared error between predicted and true valence/arousal, and $\alpha = 0.5$ in our experiments.

D. Task 4: Contrastive Text-Audio Alignment

Task 4 trains a dual-encoder model to align text and audio in a shared embedding space. The BERT and GNN encoders from Tasks 1 and 2 are reused and each followed by a small linear projection head. For a batch of tracks, we compute the cosine similarity between every text projection and every graph projection, and train with a symmetric InfoNCE contrastive loss so that the embeddings of a matching (audio, caption) pair are pulled together while all other pairs in the batch are pushed apart. The intended evaluation for this task is recall-at- K : whether the correct match for a given query appears among the top K most similar items. As described in Section VI, we were not able to obtain results for this task due to a data-loading issue and report the method only.

V. IMPLEMENTATION DETAILS

The full pipeline is implemented in a single Jupyter notebook using PyTorch and the Hugging Face `transformers` library, run on Google Colab. Table I summarizes the training configuration for each task. All models were trained with the Adam optimizer.

TABLE I
TRAINING CONFIGURATION BY TASK

Task	Epochs	Learning rate	Hidden dim.
Task 1 (BERT)	60	1×10^{-2}	768 (BERT)
Task 2 (GNN)	25	1×10^{-3}	32
Task 3 (Fusion)	25	1×10^{-3}	64
Task 4 (Contrastive)	20	1×10^{-3}	32 (projection)

VI. RESULTS

Table II reports macro-F1 and area-under-precision-recall curve (AUC-PR) for the classification tasks, evaluated on a held-out 20% split of the DEAM tracks. Table III reports the emotion regression results for the Task 3 fusion model. Fig. 1 shows training loss curves for Task 1 and Task 2, and Fig. 2 shows a t-SNE projection of the fused embeddings from Task 3, colored by mood-quadrant label.

As Table II shows, both the BERT-only baseline and both fusion variants reach a macro-F1 and AUC-PR of 1.000. The GNN-only model clearly outperforms the CNN baseline in macro-F1 (0.296 versus 0.258), though both reach a similar AUC-PR. This suggests that the graph structure does provide some benefit over a model with no relational information,

TABLE II
CLASSIFICATION RESULTS ON THE DEAM HELD-OUT TEST SET

Model	Macro F1	AUC-PR
Random baseline	~0.05	~0.15
CNN baseline (no graph)	0.258	0.384
BERT only (Task 1)	1.000	1.000
GNN only (Task 2)	0.296	0.384
Fusion, concatenation (Task 3)	1.000	1.000
Fusion, attention (Task 3)	1.000	1.000

TABLE III
EMOTION REGRESSION RESULTS (TASK 3 FUSION, CONCATENATION VARIANT)

Target	MAE	R^2
Valence & arousal (mean)	0.114	0.107

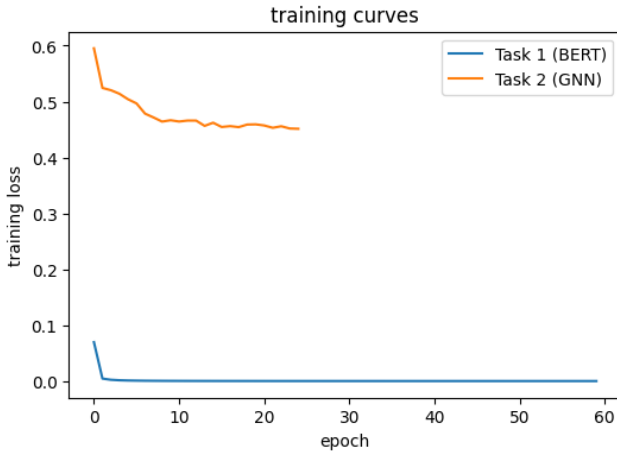


Fig. 1. Training loss curves for Task 1 (BERT) and Task 2 (GNN).

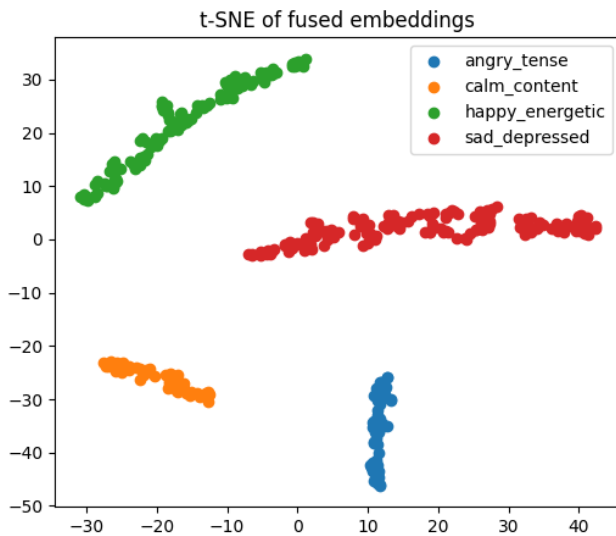


Fig. 2. t-SNE projection of the fused Task 3 embeddings on the test set, colored by mood-quadrant label.

though the GNN-only model is still far from perfect accuracy when using audio alone. The emotion regression results in Table III show a low R^2 of 0.107, indicating that predicting continuous valence and arousal is a substantially harder problem for our fusion model than predicting the four discrete mood quadrants. The t-SNE plot in Fig. 2 shows four visually distinct clusters corresponding to the four mood-quadrant labels, consistent with the near-perfect classification scores.

VII. LIMITATIONS AND FUTURE WORK

Two limitations should be noted when interpreting the results above.

Label leakage in the text pathway. The pseudo-captions used for Task 1 and Task 3 are constructed directly from the same mood-quadrant label being predicted (Section II-A). Because of this, the text pathway can solve the classification task almost trivially, which is the most likely explanation for the perfect scores obtained by the BERT-only model and both fusion variants. This means Table II does not yet show a fair comparison between the fusion model and the text-only baseline, since the fusion model’s classification head can rely entirely on the text pathway and ignore the graph pathway. A direct test of the graph’s added value should instead compare against the GNN-only row, which does not have this shortcut available. Future work should evaluate the fusion model using text that does not directly encode the label, such as free-text captions from MusicCaps, to obtain a meaningful fusion-versus-baseline comparison.

Task 4 data loading. At the time of writing, our FMA-small and MusicCaps loaders returned zero usable tracks, which we traced to metadata and audio-path mismatches introduced while adapting the loaders to our Colab environment. As a result, Task 4 fell back to training on an unintended subset of data and its contrastive loss and recall-at- K results are not reported here. We plan to correct the dataset paths and re-run Task 4 before final submission.

VIII. CONCLUSION

We implemented all four tasks specified in the assignment: a BERT text baseline, a GraphSAGE-style GNN over audio segment graphs, a GNN-BERT fusion model with a multi-task classification and emotion-regression objective, and a contrastive text-audio alignment model. Using DEAM as our primary dataset, and MusicCaps combined with FMA-small for the advanced task, we obtained classification and regression results for Tasks 1 through 3, and identified two concrete issues, a label-leakage effect in the text pathway and an unresolved data-loading bug in Task 4, that we intend to address before final submission.

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